Date. No. 3-DYLIK MODELBUILDING_204LIK

(Visualization kutubxonalari)

class = f1 + f2 + f3 ...

Visualization kutubxonalar

- Matplotlip
- Seaborn
- Plotly Express

-Matplotlip

~Sodda, statik visualization uchun ishlatiladigan Python kutubxonasi

~ Quyidagi xollarda ishlaydi:

- 1ta yoki 2ta input qiymatlar va target qiymat ortasidagi munosabatni korish uchun
- Istalgan ikki input qiymatlarni orasidagi munosabatlarni korish uchun ishlatiliadi

Matplotlip → (#1chi Missing valueni toldirish kerak)

```
import matplotlip.pyplot as plt
df.info()
#Age Distribution (yosh taqsimoti)
df['age'].value_counts()
plt.figure(figsize=(10,6)) #Figura olchami
plt.hist(df['age']) => plt.hist(df['age'], bins=20, color='red')
                                #Figura turi histogram
plt.show()
plt.hist(df['age'], bins=20, color='skyblue', edgecolor='black')
plt.show
ptt.title('Yosh taqsimoti')
```

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Matplotlib amaliy

```
import matplotlib.pyplot as plt
# matplotlib u/n missing value muhim (oldin missing value qismi kerak)
# Age -> Distribution (yosh tagsimoti)
df['age'].value_counts()
plt.figure (figsize = (10,6))    # Figure o'lchami
plt.hist(df['age'], bins=20, color='skyblue', edgecolor='black')
# Figure turi - histogram
plt.title('Yosh tagsimoti')
plt.xlabel ('Yosh')
plt.ylabel ('Soni')
plt.show()
#Matplotlib da koprog classlari koprogini olish kerak
# pychart -> classlar kam bolsa
```

Seaborn

• Matplotlibning ustiga qurilgan chiroyli va koprog ma'lumot beradigan Python kutubxonasi

Seaborn amaliy

```
df['chol'].value_counts()
#Gender bo'yicha xolisetrin (chol) darajasini topish tagsimoti
import seaborn as sns
① plt.figure(figsize=(10,8))
sns.boxplot (x='gender', y='chol', data=df)
plt.show()
#Biz Seabornni ishlatish u/n albatta Matplotlibni import qilgan bo'lishimiz k/k
```

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②

```
plt.figure(figsize=(10,8))
sns.boxen(x='gender', y='chol', data=df)
plt.show()
```

boxenplot

⇓

```
plt.figure(figure=(10,8))
sns.boxenplot(x='gender', y='chol', data=df)
plt.show()
```

Seaborn ⇒ x labelda classlar ko'p bolsa yaxshi natija ola olmaymiz

Plotly express

- Qiziqarli va boshqariluvchan visual chizmalar uchun ishlatiladi. ((Yaqinlashtirish, Uzoqlashtirish, Filter qilish))

Plotly express amaliy (practical) bilan ishlatish

```
import plotly.express as px
fig = px.scatter(df, x='age', y='thalch', color='num', size='chol',
hover_data=['gender', 'trestbps'], title='Qon bosimini target qiymat b/n aloqasi')
```

Plotly express

Rang color='Columns name' ⇒ Nuqtalar rangi 'Columns' ga qiymatiga qarab o'zgaradi

Nuqta o'lchami size='Columns' ⇒ Nuqtaning kattaligi ham 'columns' ga bog'liq bo'ladi

Hover data hover_data=['Columns'] ⇒ Sichqonchani nuqta ustiga olib borsangiz qo'shimcha info chiqadi

Umumiy xulosa:

- Asosiy e'tibor berilishi kerak bo'lgan xolatlar
- Missing valuelardan tozalash kerak
- Tanlab olinayotgan input qiymatlar chizmalarga mos bo'lishi kerak.

Kutubxona_to`g`irlangani.ipynb

Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline

```
df_1000 = df.sample(n=1000) # Data ko'pligi uchun 1000 ta malumotnigina oldik
```

✓ 0.0s

```
fig=px.scatter(df_1000,x='Daily_Phone_Hours',y='Stress_Level',color='Stress_Level',size='Social_Media_Hours',hover_data=['Stress_Level','Daily_Phone_Hou
print(fig.show())
```

✓ 0.7s